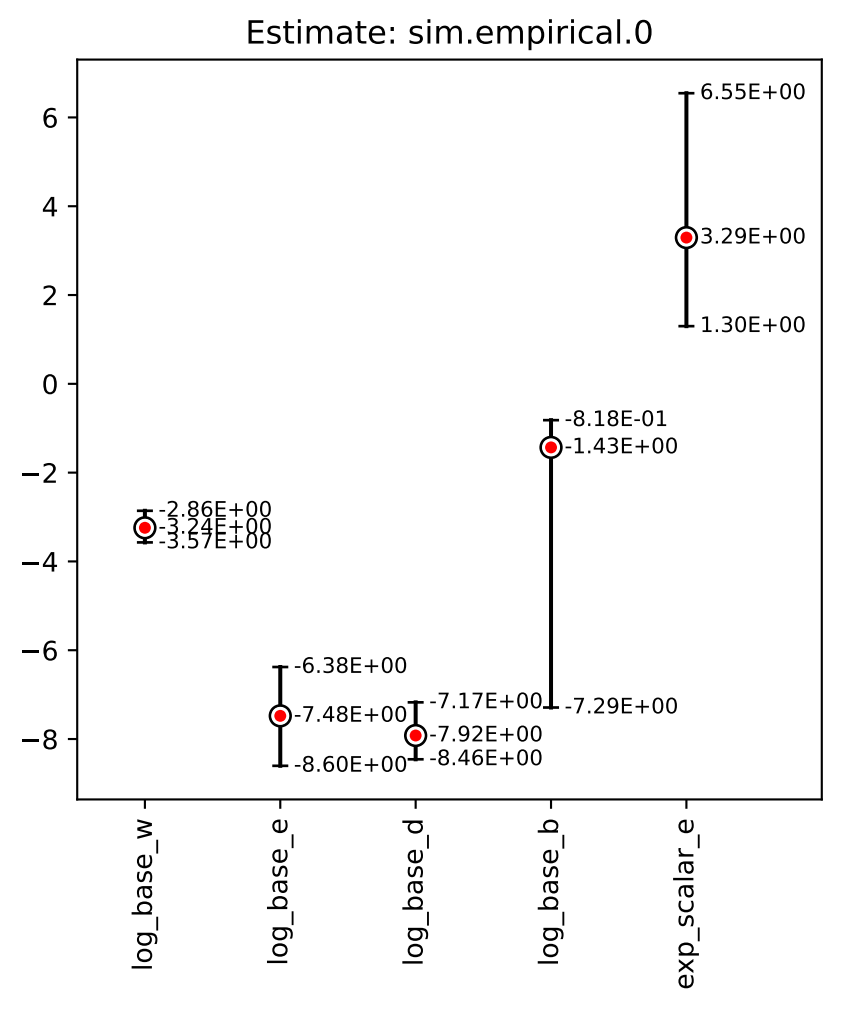
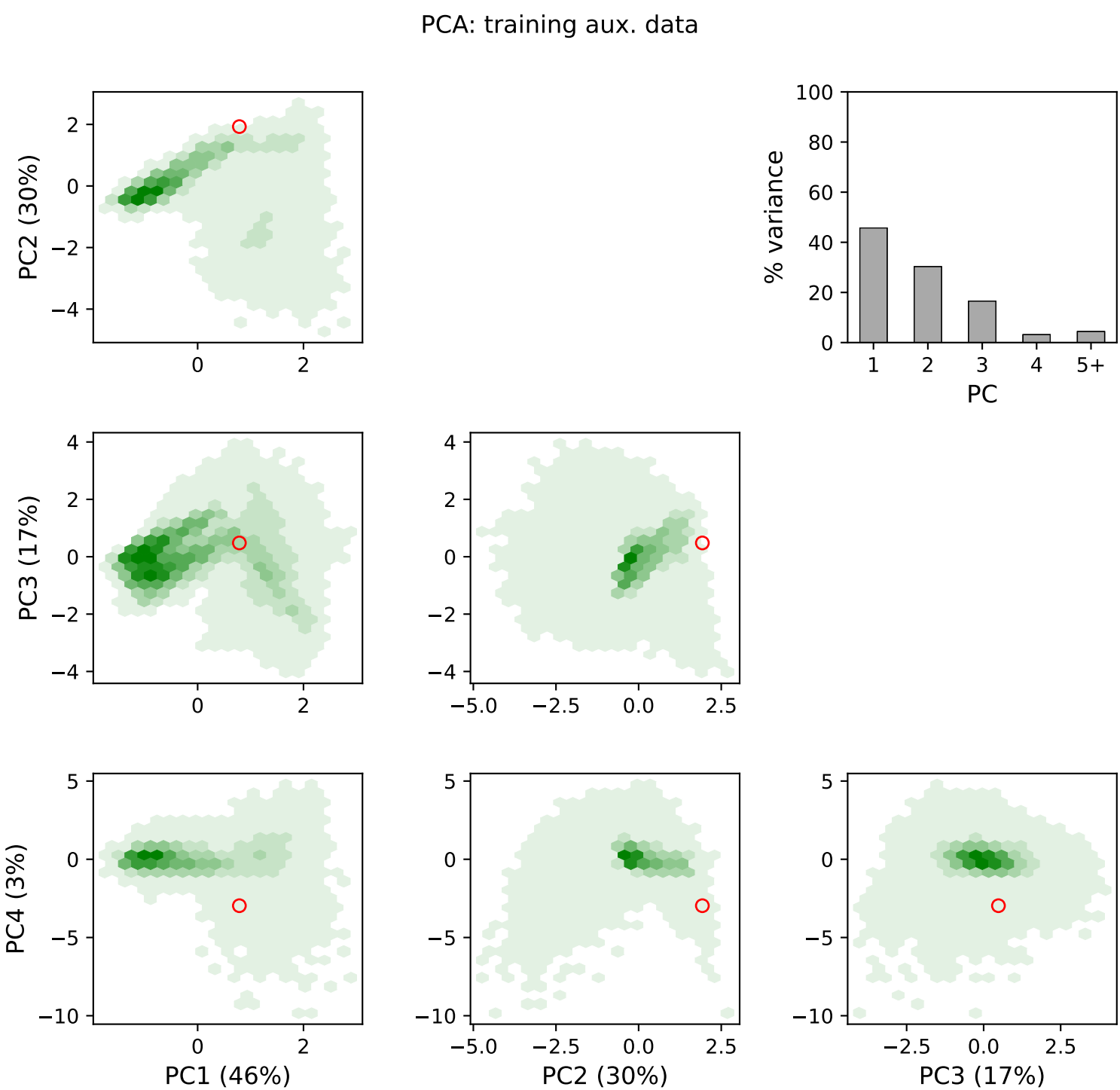
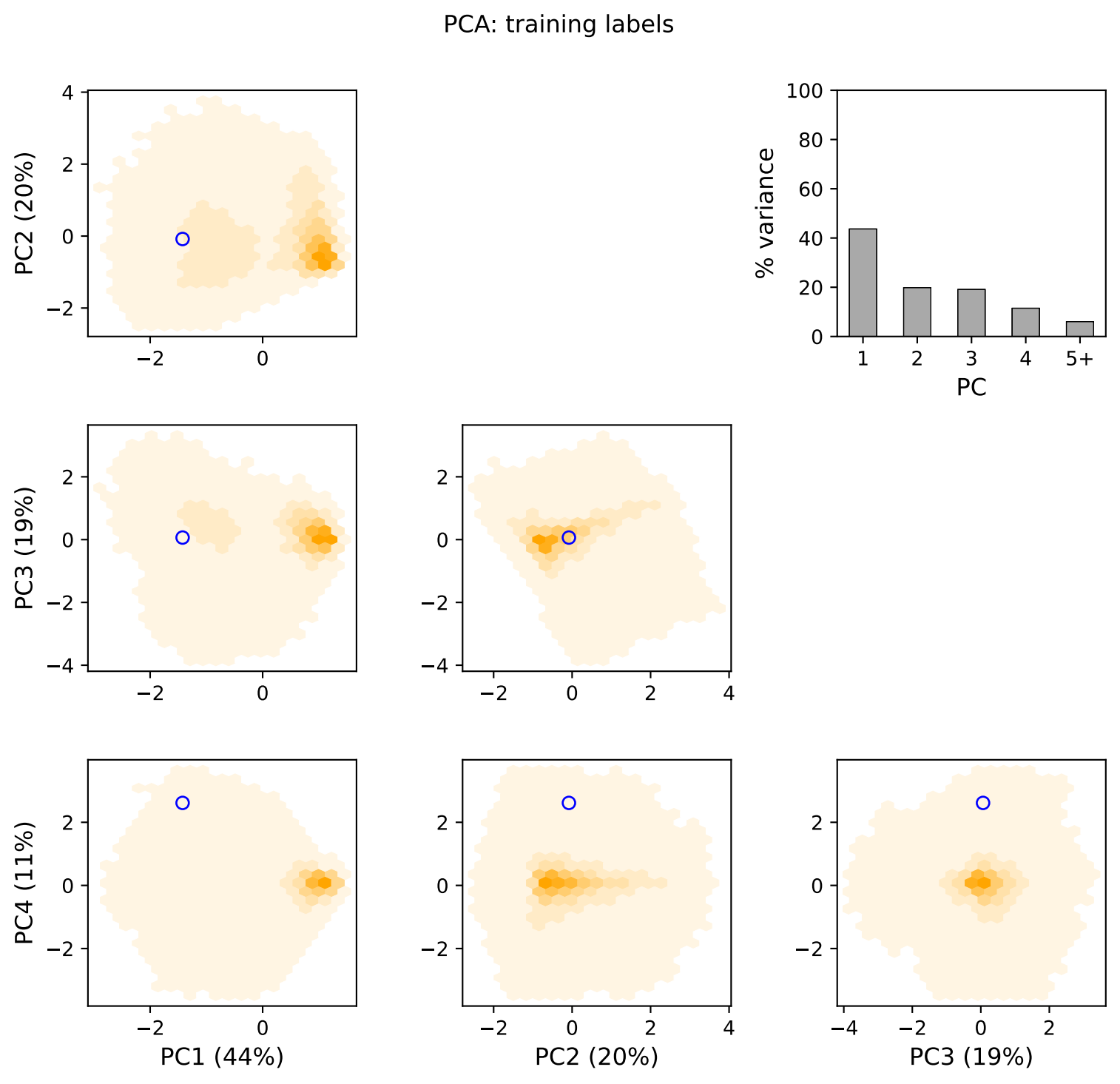
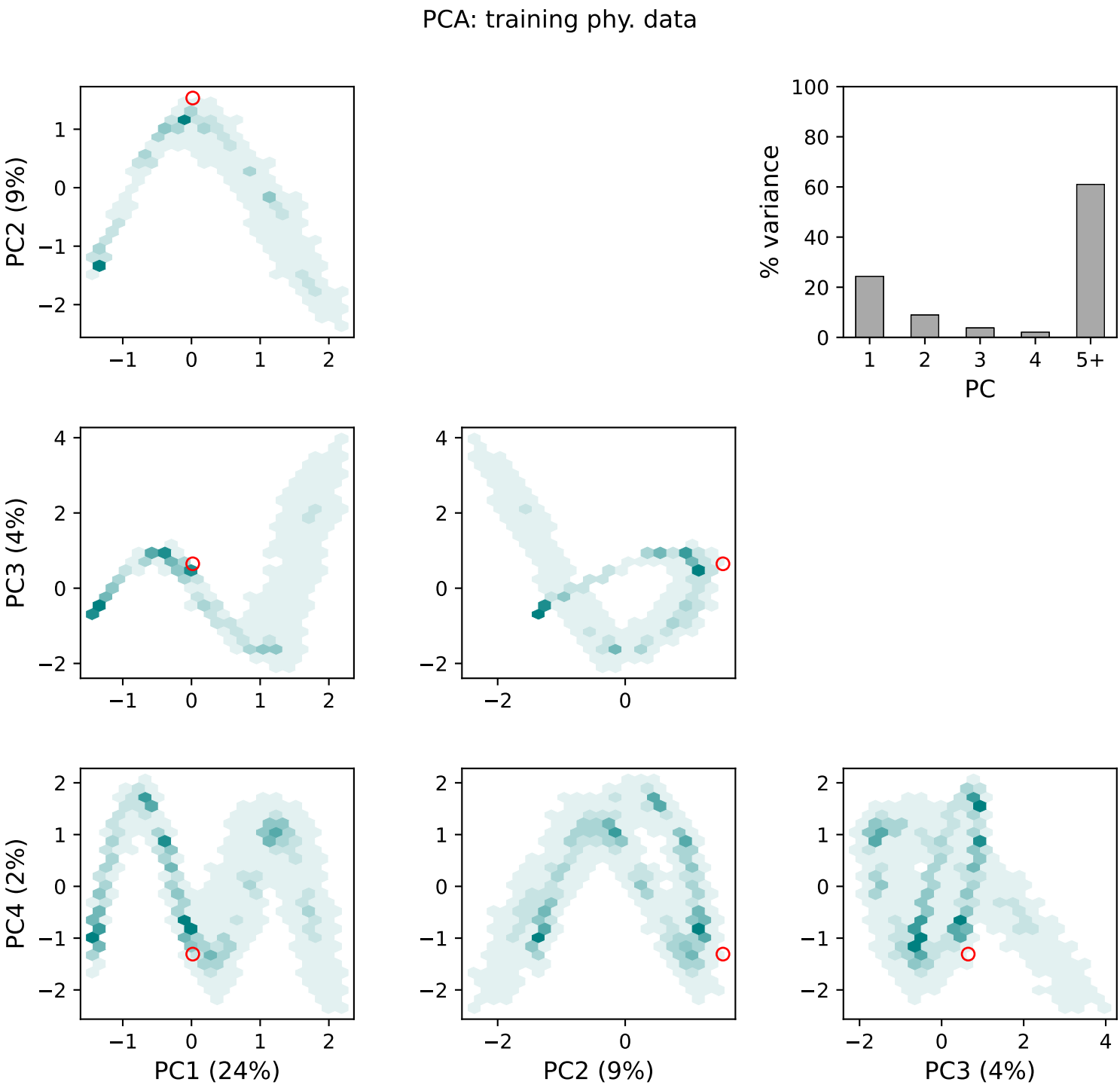
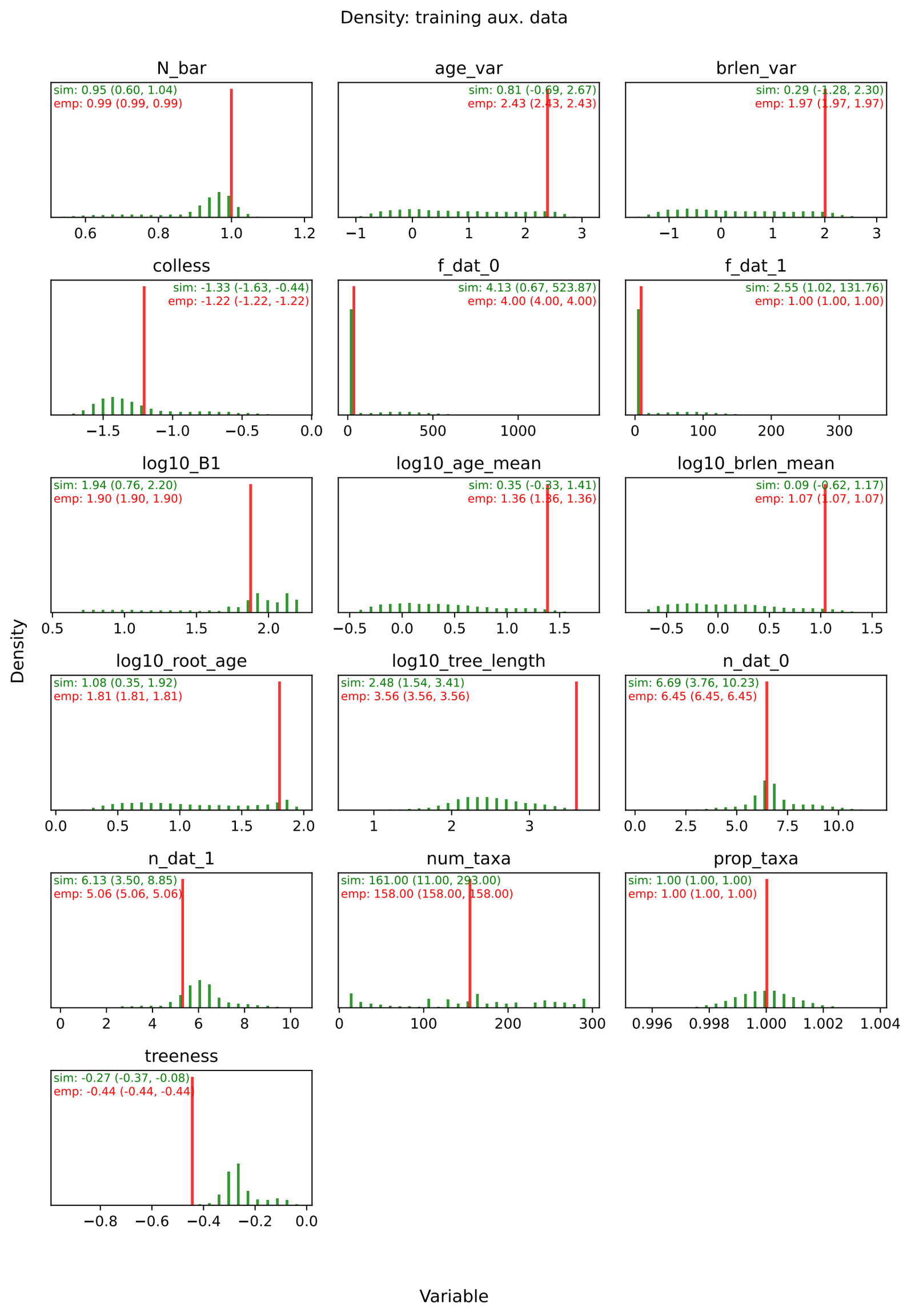
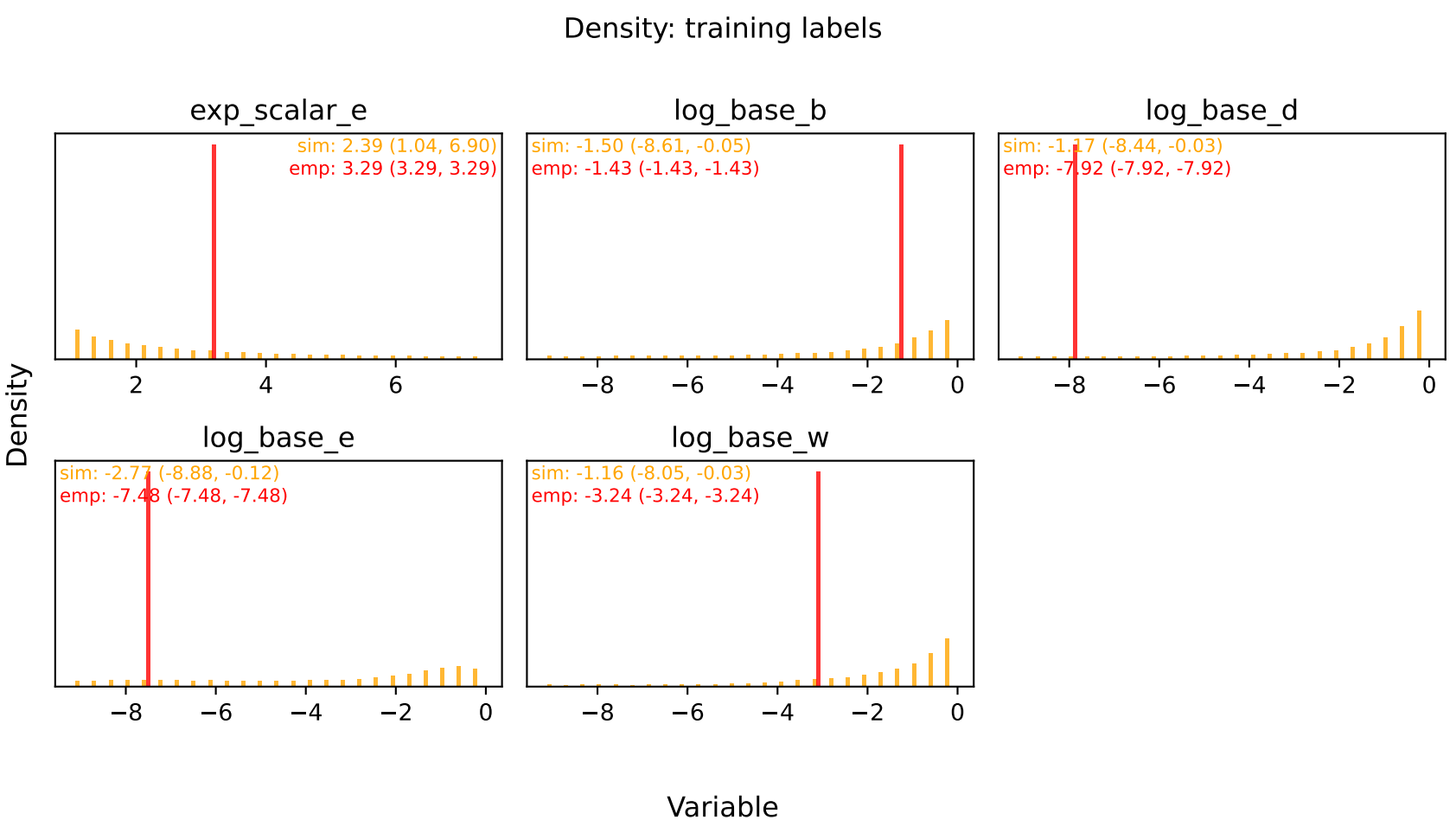

Estimate: sim.empirical.0
6.55E+00
3.29E+00
1.30E+00
-8.18E-01
-1.43E+00
-2.86E+00
-3.24E+00
-3.57E+00
-6.38E+00
-7.48E+00
-8.60E+00
-7.17E+00
-7.92E+00
-8.46E+00
-7.29E+00
6
4
2
0
−2
−4
−6
−8
log_base_w
log_base_e
log_base_d
log_base_b
exp_scalar_e

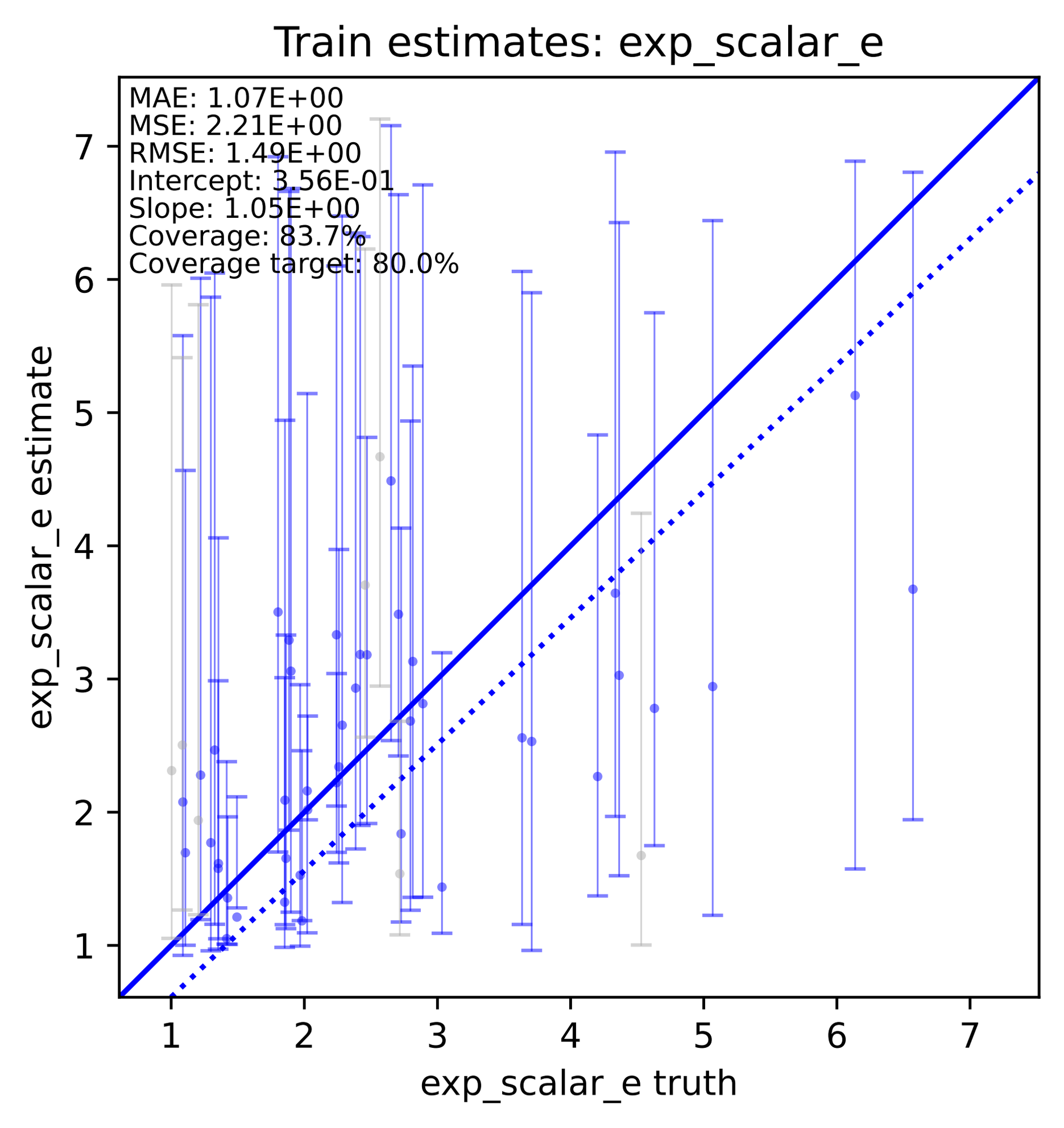

PCA: training aux. data
PC2 (30%)
% variance
PC
PC3 (17%)
PC4 (3%)
PC1 (46%)
PC2 (30%)
PC3 (17%)

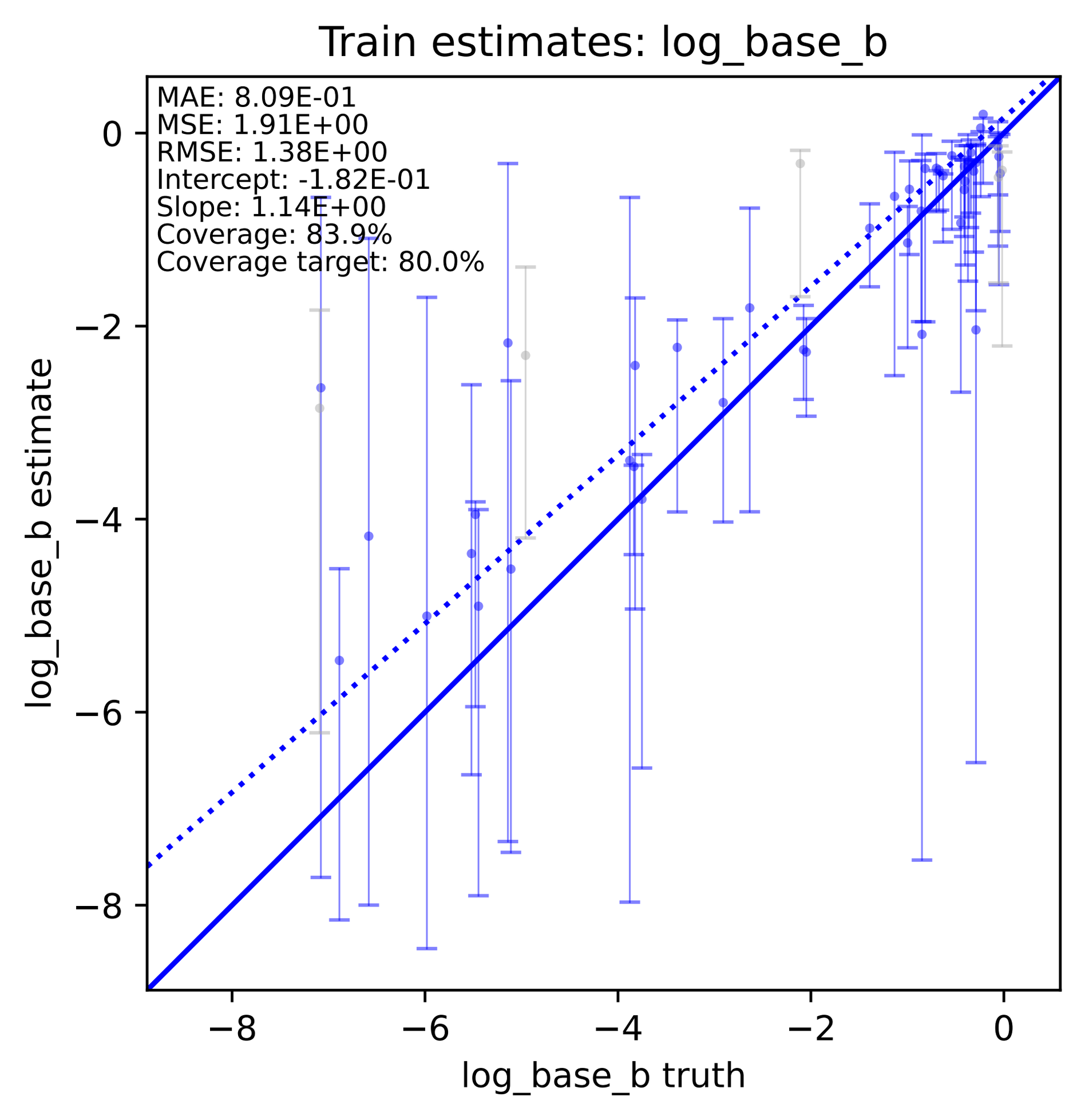

PCA: training labels
PC2 (20%)
% variance
PC
PC3 (19%)
PC4 (11%)
PC1 (44%)
PC2 (20%)
PC3 (19%)

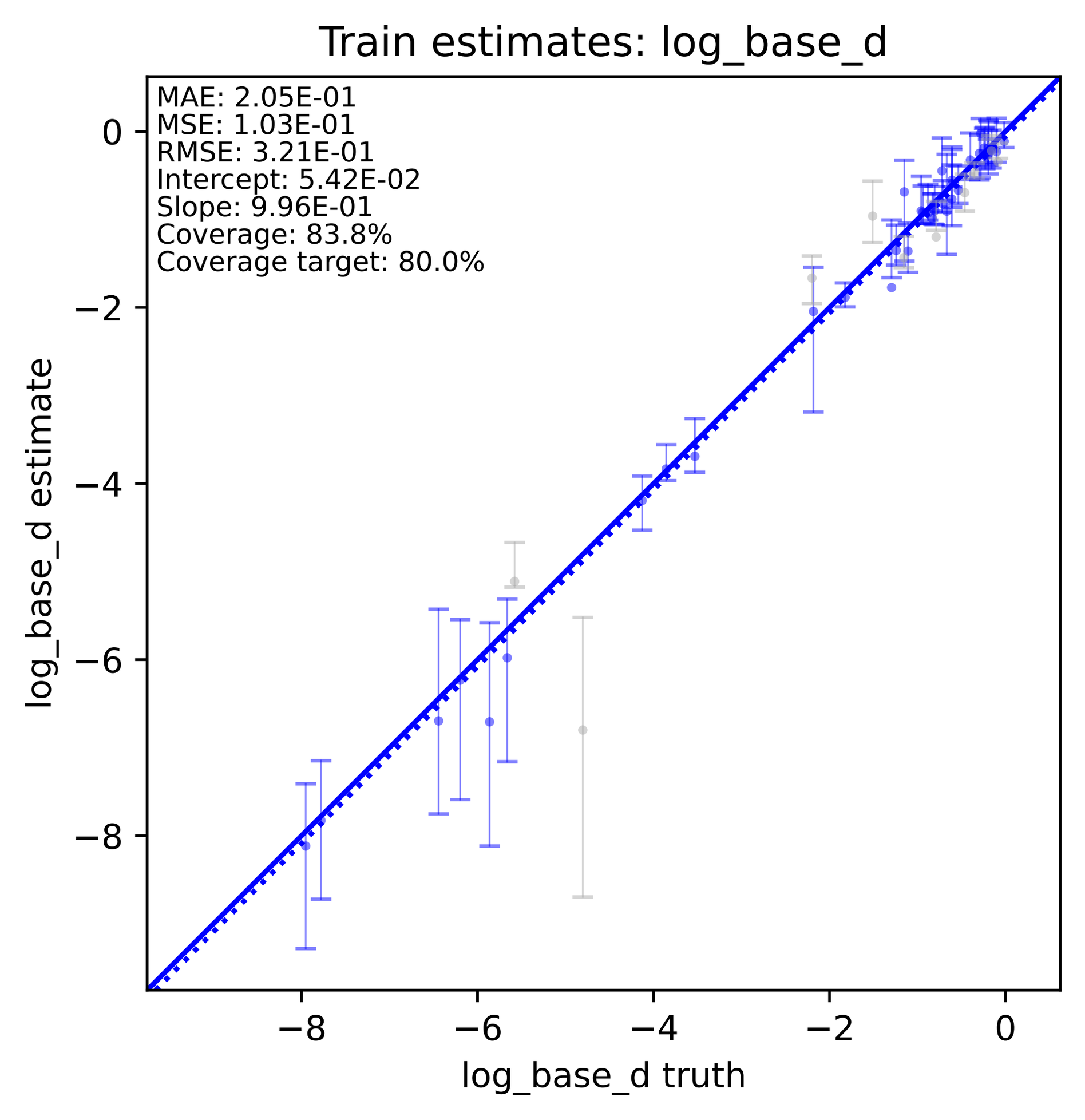

PCA: training phy. data
PC2 (9%)
% variance
PC
PC3 (4%)
PC4 (2%)
PC1 (24%)
PC2 (9%)
PC3 (4%)

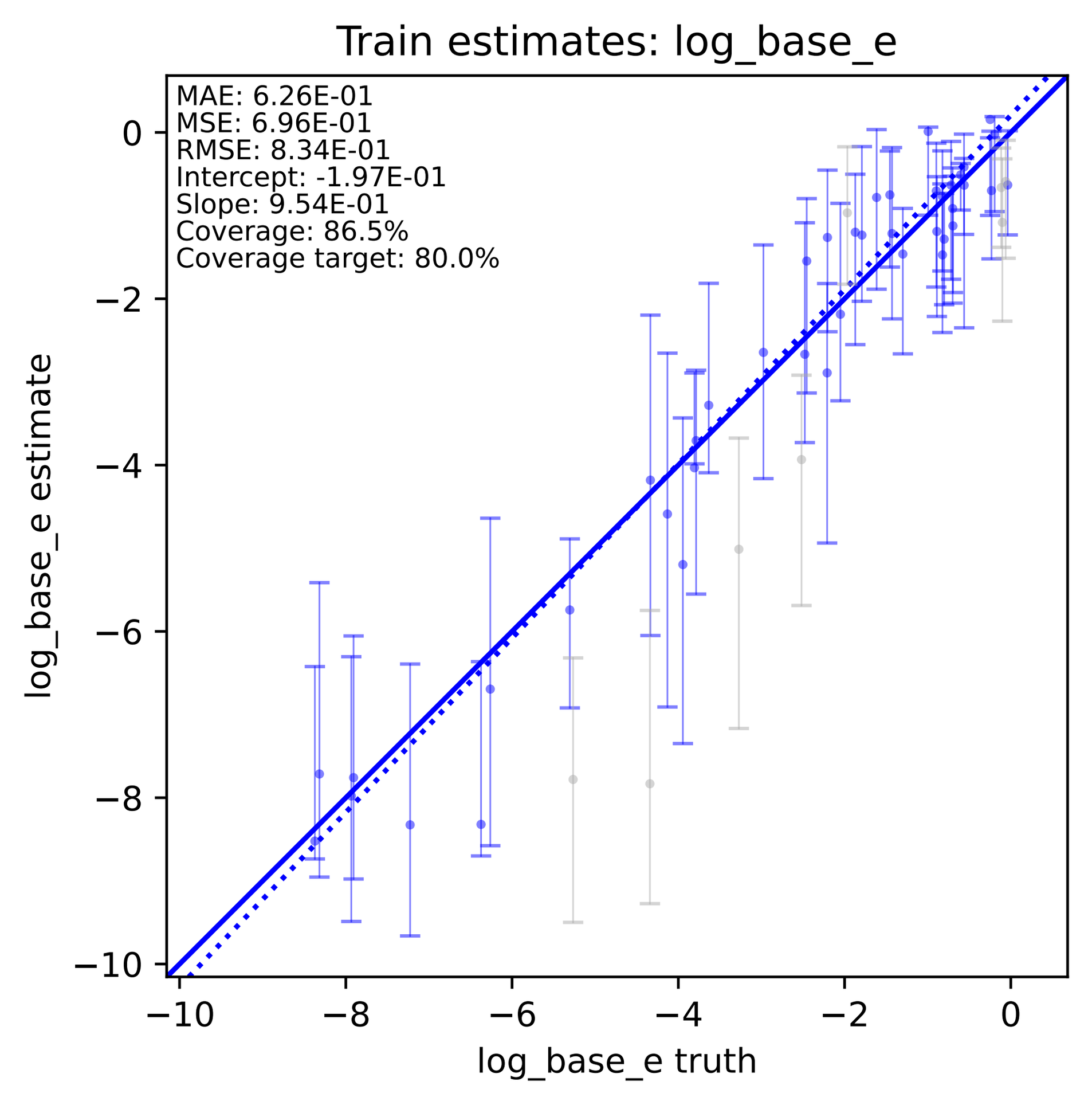

Density: training aux. data
N_bar
sim: 0.95 (0.60, 1.04)
emp: 0.99 (0.99, 0.99)
age_var
sim: 0.81 (-0.69, 2.67)
emp: 2.43 (2.43, 2.43)
brlen_var
sim: 0.29 (-1.28, 2.30)
emp: 1.97 (1.97, 1.97)
colless
sim: -1.33 (-1.63, -0.44)
emp: -1.22 (-1.22, -1.22)
f_dat_0
sim: 4.13 (0.67, 523.87)
emp: 4.00 (4.00, 4.00)
f_dat_1
sim: 2.55 (1.02, 131.76)
emp: 1.00 (1.00, 1.00)
log10_B1
sim: 1.94 (0.76, 2.20)
emp: 1.90 (1.90, 1.90)
log10_age_mean
sim: 0.35 (-0.33, 1.41)
emp: 1.36 (1.36, 1.36)
log10_brlen_mean
sim: 0.09 (-0.62, 1.17)
emp: 1.07 (1.07, 1.07)
log10_root_age
sim: 1.08 (0.35, 1.92)
emp: 1.81 (1.81, 1.81)
log10_tree_length
sim: 2.48 (1.54, 3.41)
emp: 3.56 (3.56, 3.56)
n_dat_0
sim: 6.69 (3.76, 10.23)
emp: 6.45 (6.45, 6.45)
n_dat_1
sim: 6.13 (3.50, 8.85)
emp: 5.06 (5.06, 5.06)
num_taxa
sim: 161.00 (11.00, 293.00)
emp: 158.00 (158.00, 158.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.27 (-0.37, -0.08)
emp: -0.44 (-0.44, -0.44)
Density
Variable

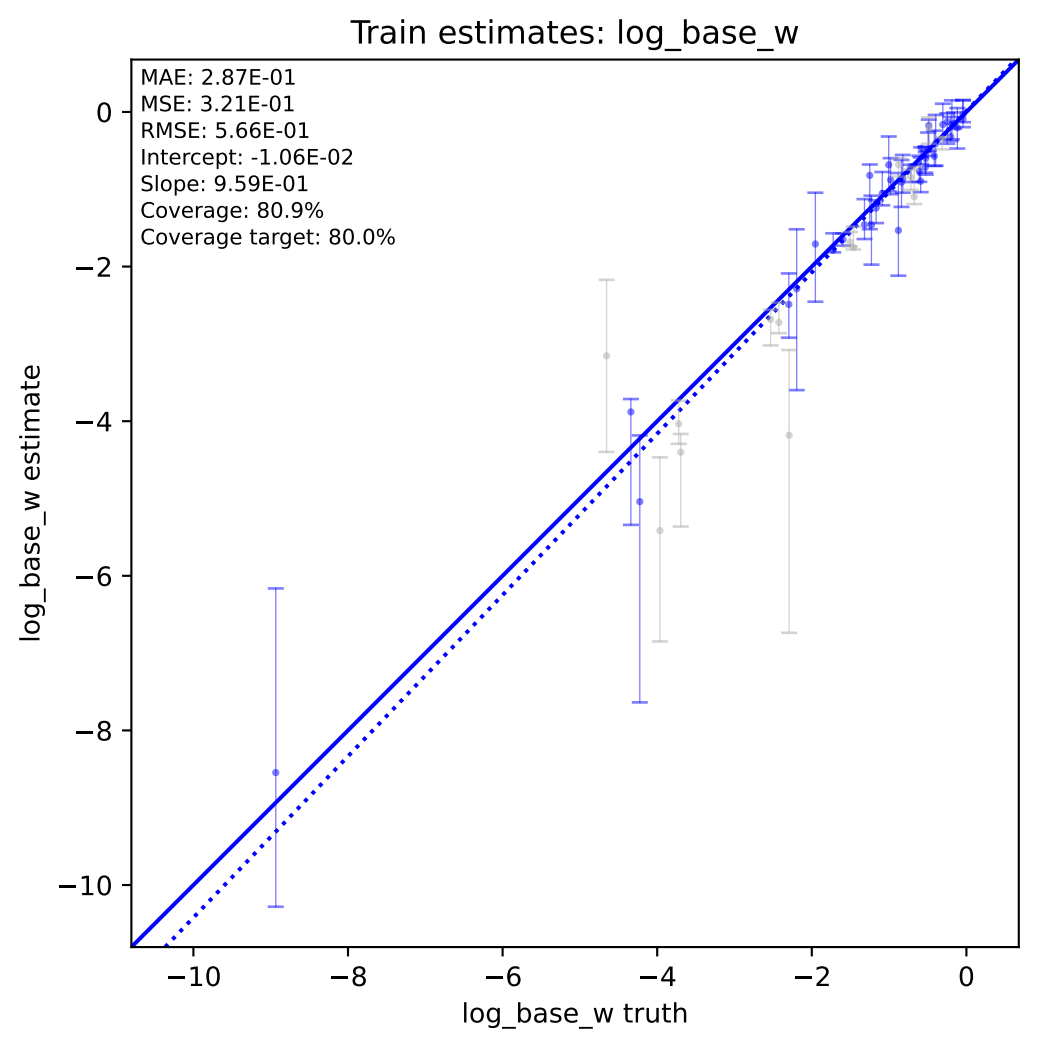

Density: training labels
exp_scalar_e
sim: 2.39 (1.04, 6.90)
emp: 3.29 (3.29, 3.29)
log_base_b
sim: -1.50 (-8.61, -0.05)
emp: -1.43 (-1.43, -1.43)
log_base_d
sim: -1.17 (-8.44, -0.03)
emp: -7.92 (-7.92, -7.92)
log_base_e
sim: -2.77 (-8.88, -0.12)
emp: -7.48 (-7.48, -7.48)
log_base_w
sim: -1.16 (-8.05, -0.03)
emp: -3.24 (-3.24, -3.24)
Density
Variable

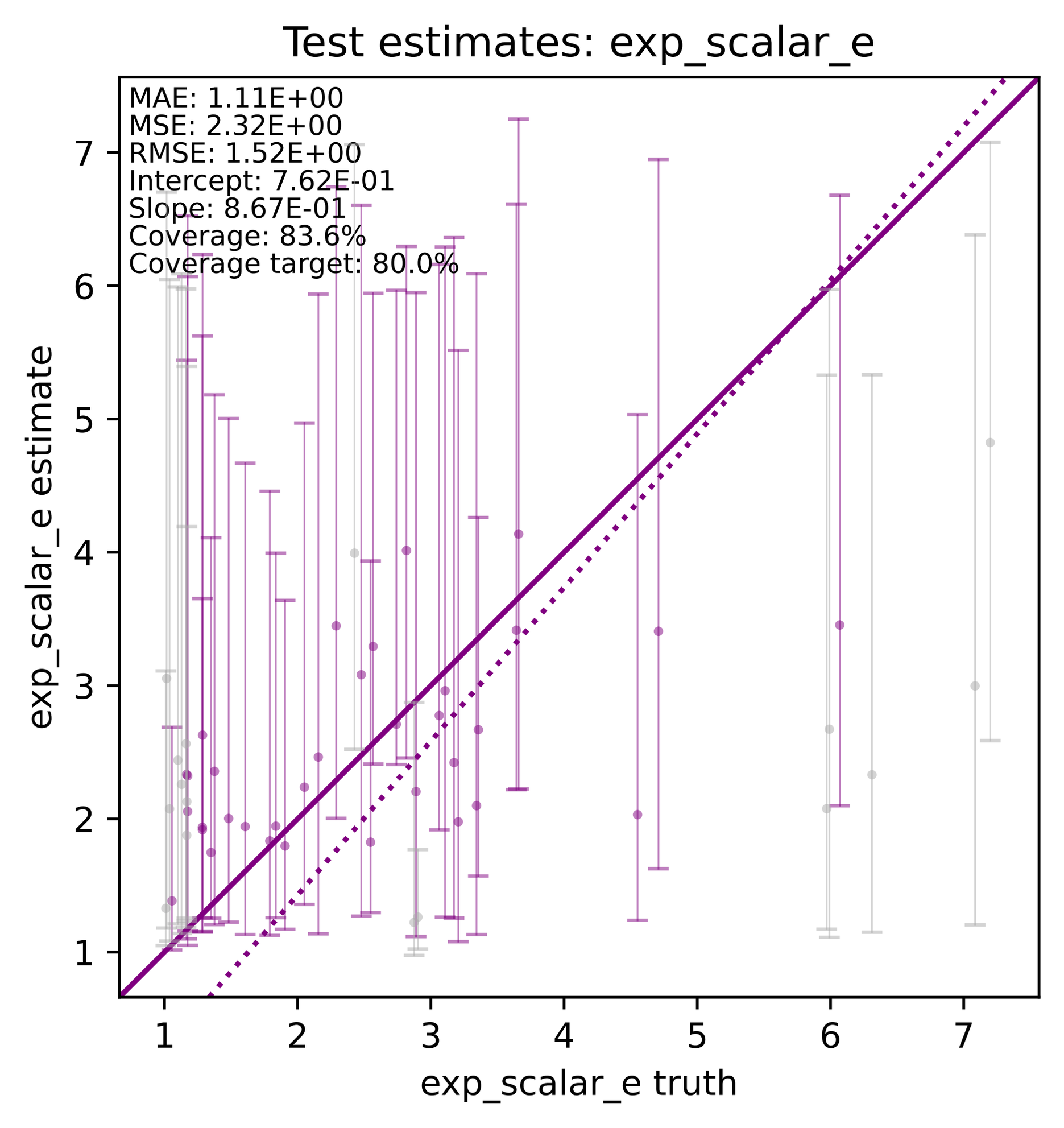

Train estimates: exp_scalar_e
MAE: 1.07E+00
MSE: 2.21E+00
RMSE: 1.49E+00
Intercept: 3.56E-01
Slope: 1.05E+00
Coverage: 83.7%
Coverage target: 80.0%
exp_scalar_e estimate
exp_scalar_e truth
1
2
3
4
5
6
7

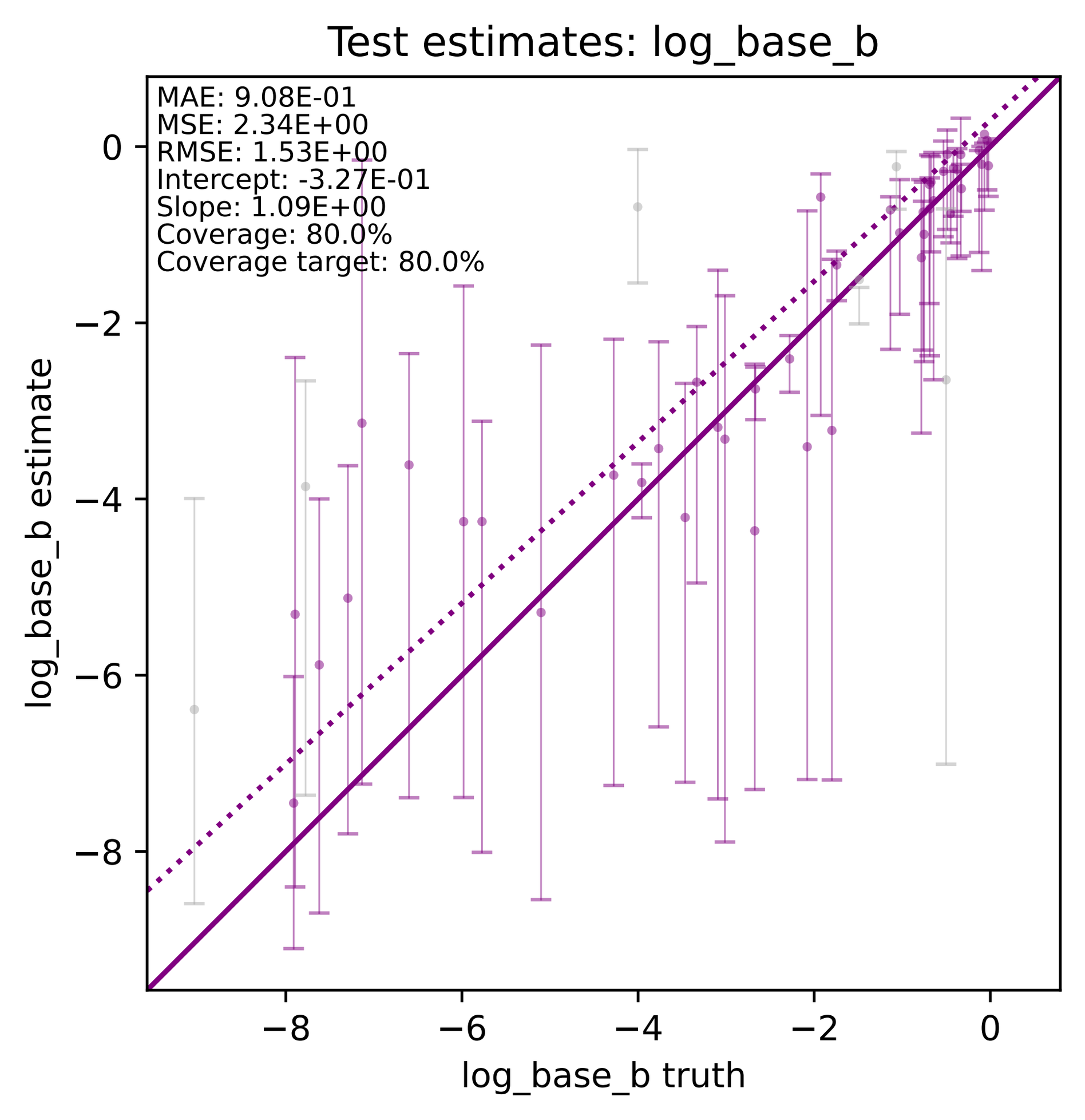

Train estimates: log_base_b
MAE: 8.09E-01
MSE: 1.91E+00
RMSE: 1.38E+00
Intercept: -1.82E-01
Slope: 1.14E+00
Coverage: 83.9%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

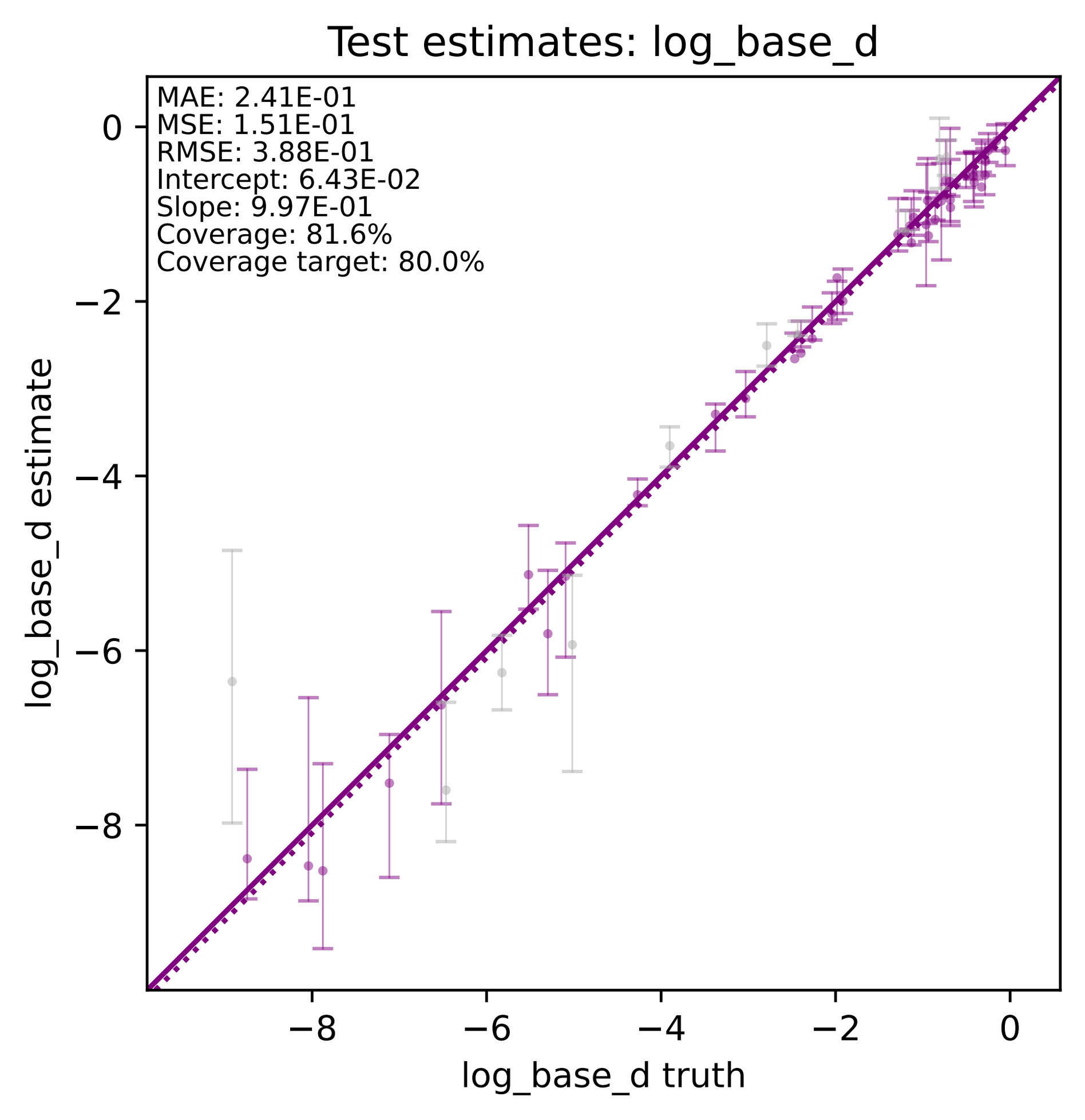

Train estimates: log_base_d
MAE: 2.05E-01
MSE: 1.03E-01
RMSE: 3.21E-01
Intercept: 5.42E-02
Slope: 9.96E-01
Coverage: 83.8%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

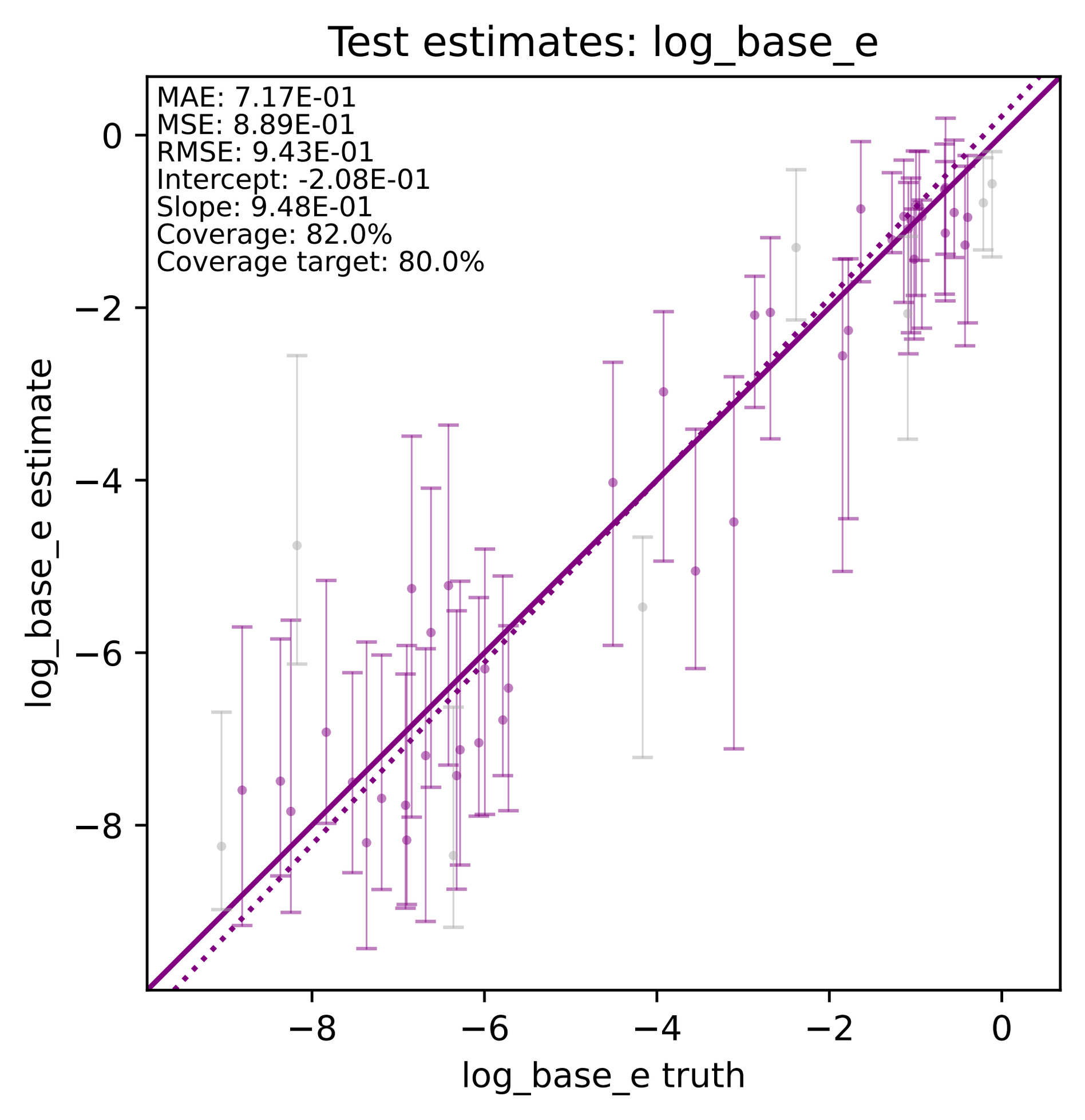

Train estimates: log_base_e
MAE: 6.26E-01
MSE: 6.96E-01
RMSE: 8.34E-01
Intercept: -1.97E-01
Slope: 9.54E-01
Coverage: 86.5%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

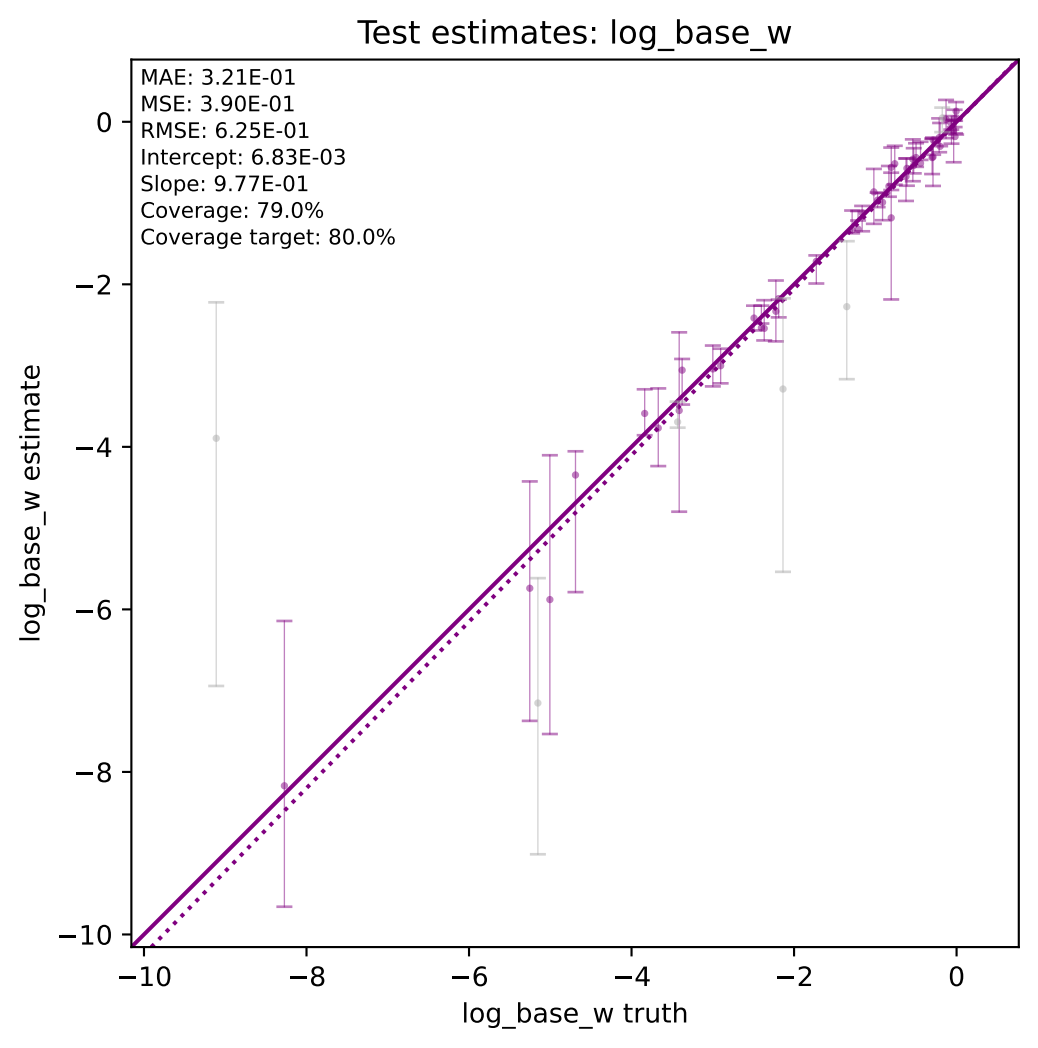

Train estimates: log_base_w
MAE: 2.87E-01
MSE: 3.21E-01
RMSE: 5.66E-01
Intercept: -1.06E-02
Slope: 9.59E-01
Coverage: 80.9%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

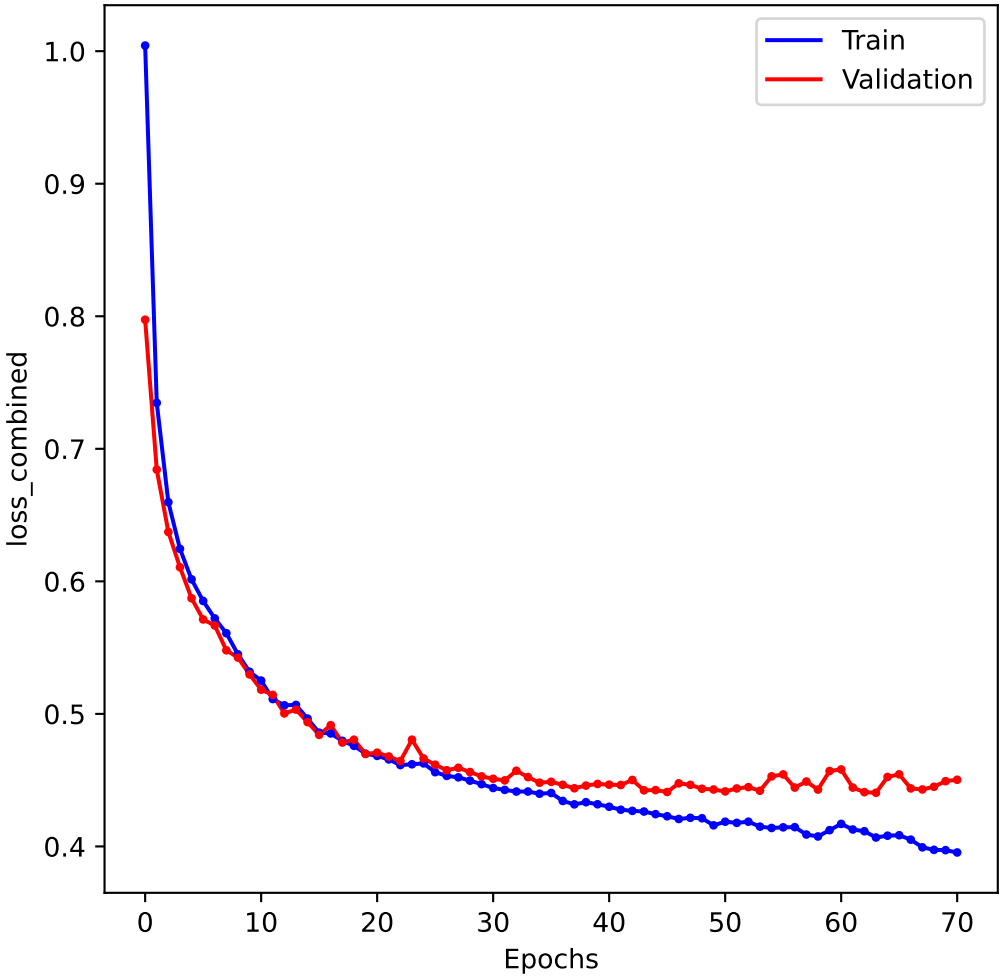

Test estimates: exp_scalar_e
MAE: 1.11E+00
MSE: 2.32E+00
RMSE: 1.52E+00
Intercept: 7.62E-01
Slope: 8.67E-01
Coverage: 83.6%
Coverage target: 80.0%
exp_scalar_e estimate
exp_scalar_e truth
1
2
3
4
5
6
7

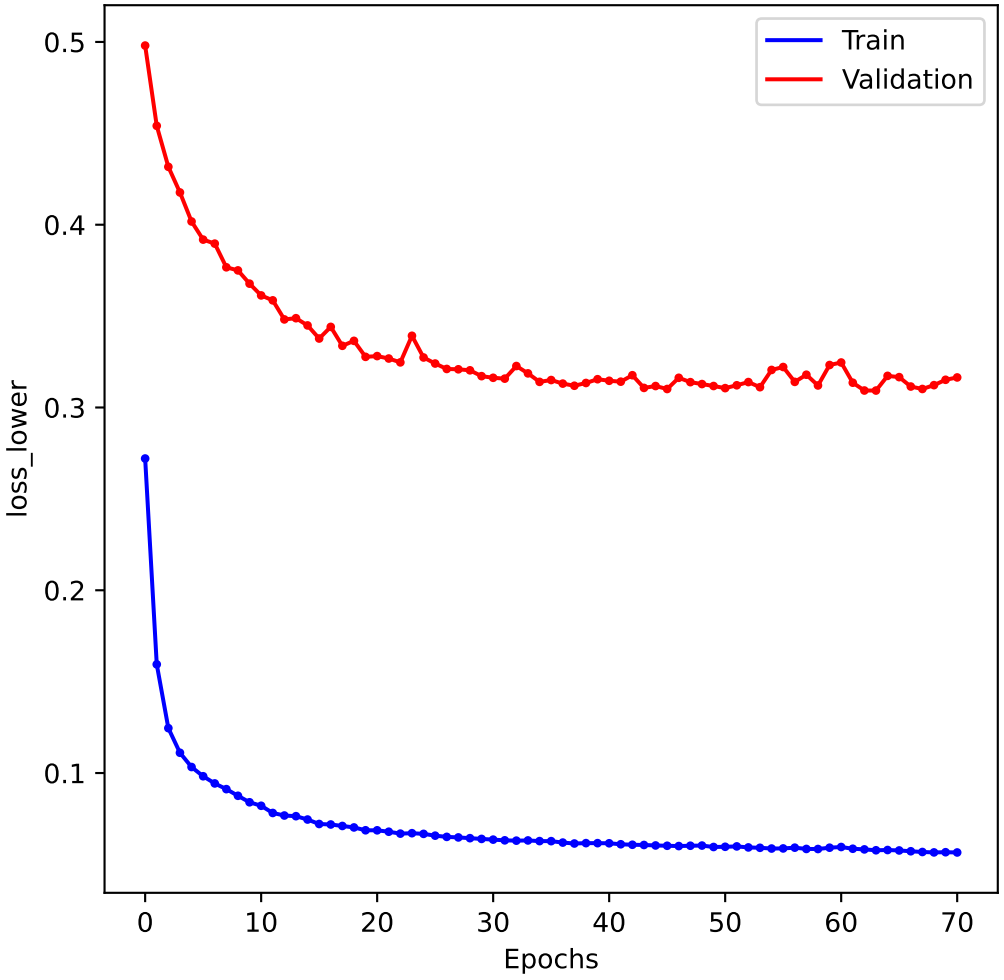

Test estimates: log_base_b
MAE: 9.08E-01
MSE: 2.34E+00
RMSE: 1.53E+00
Intercept: -3.27E-01
Slope: 1.09E+00
Coverage: 80.0%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

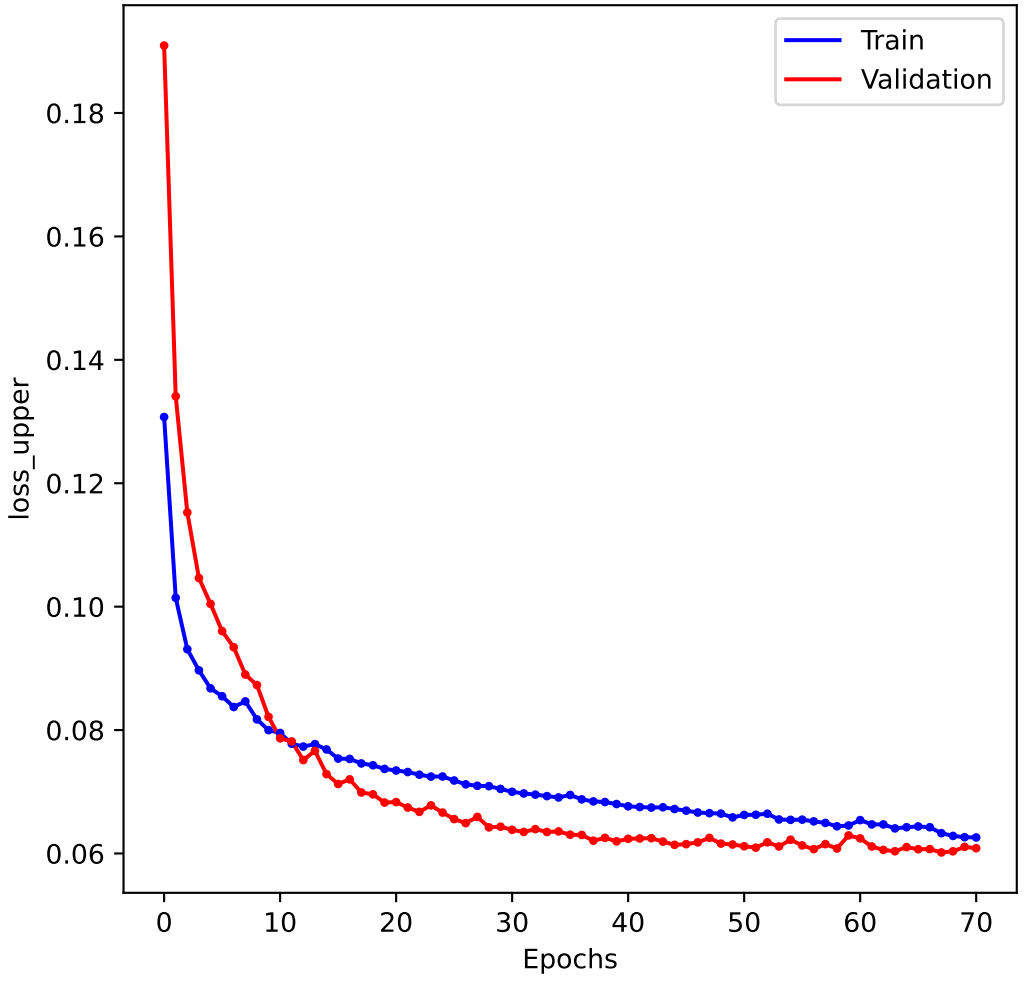

Test estimates: log_base_d
MAE: 2.41E-01
MSE: 1.51E-01
RMSE: 3.88E-01
Intercept: 6.43E-02
Slope: 9.97E-01
Coverage: 81.6%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

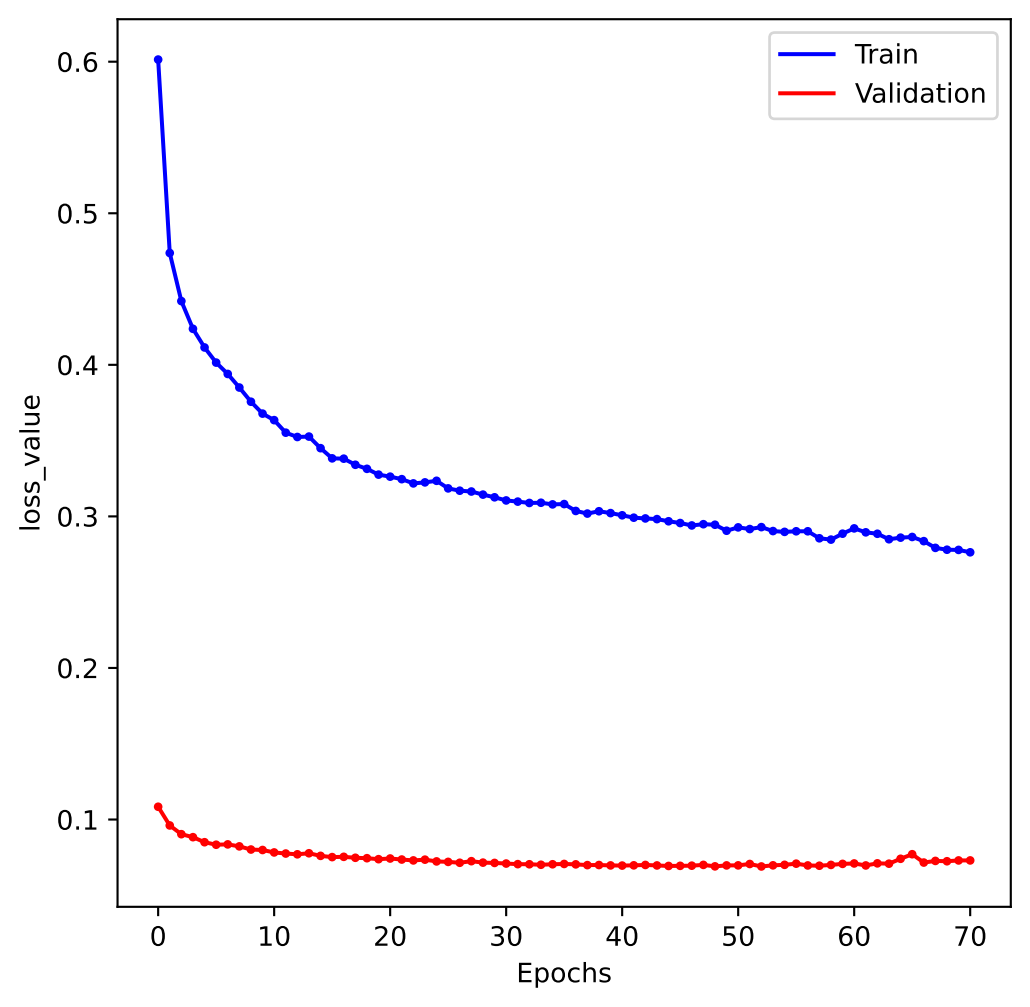

Test estimates: log_base_e
MAE: 7.17E-01
MSE: 8.89E-01
RMSE: 9.43E-01
Intercept: -2.08E-01
Slope: 9.48E-01
Coverage: 82.0%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

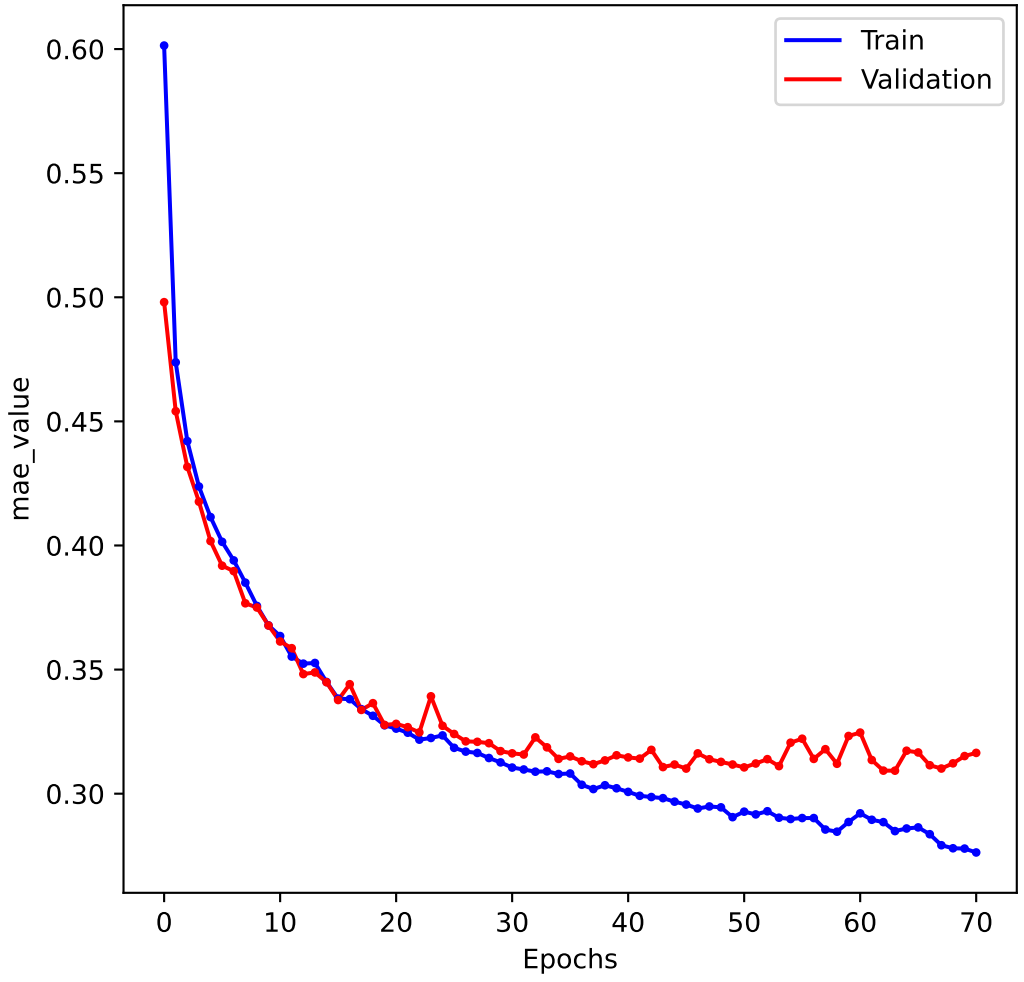

Test estimates: log_base_w
MAE: 3.21E-01
MSE: 3.90E-01
RMSE: 6.25E-01
Intercept: 6.83E-03
Slope: 9.77E-01
Coverage: 79.0%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

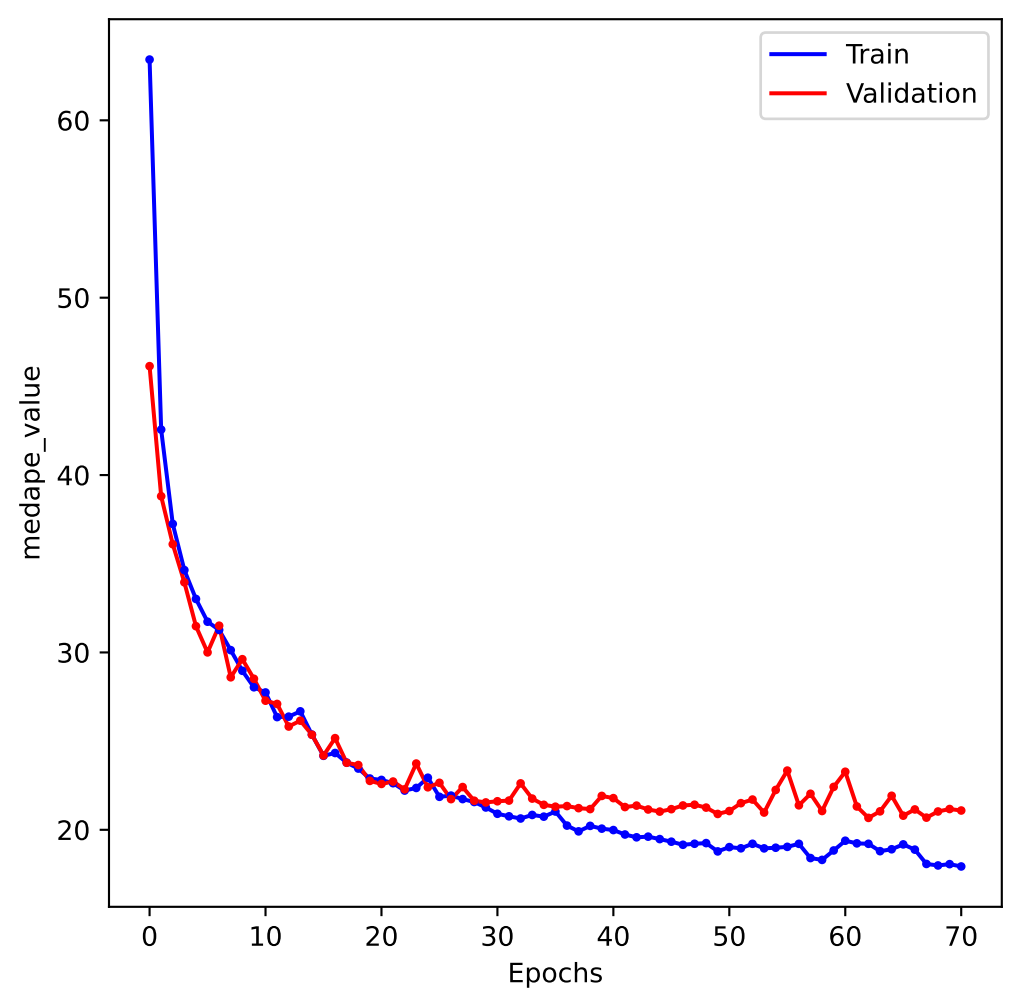

Train
Validation
loss_combined
Epochs
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0
10
20
30
40
50
60
70

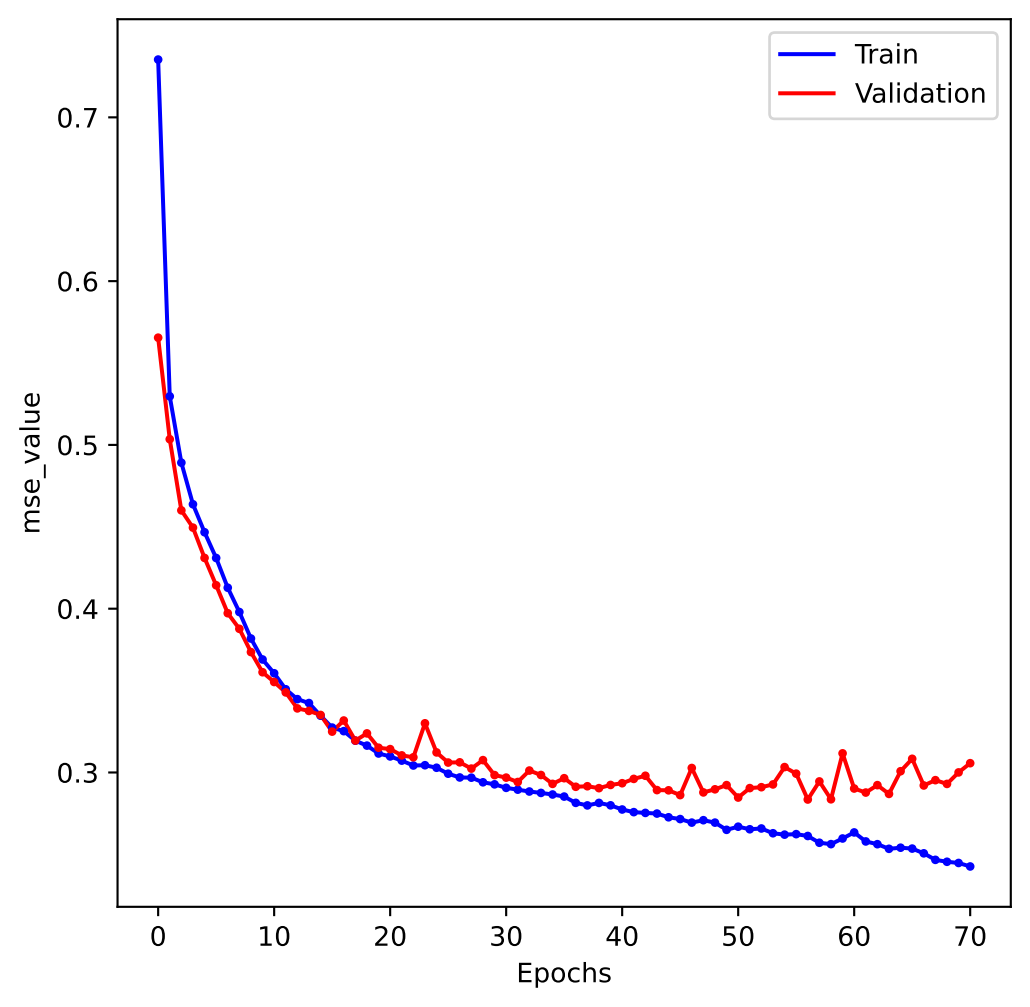

Train
Validation
loss_lower
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
50
60
70
Epochs

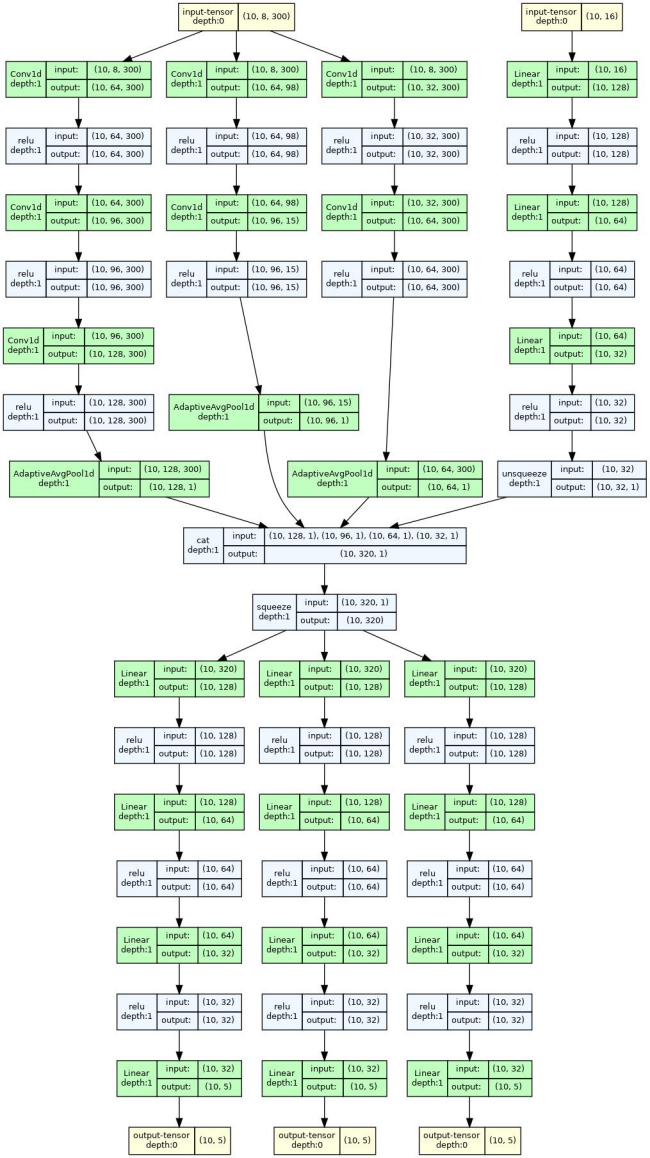

Train
Validation
loss_upper
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0
10
20
30
40
50
60
70
Epochs


Train
Validation
loss_value
0.6
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
50
60
70
Epochs


Train
Validation
mae_value
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0
10
20
30
40
50
60
70
Epochs


Train
Validation
medape_value
Epochs
60
50
40
30
20
0
10
20
30
40
50
60
70


Train
Validation
mse_value
0.7
0.6
0.5
0.4
0.3
0
10
20
30
40
50
60
70
Epochs


input-tensor depth:0
(10, 8, 300)
input-tensor depth:0
(10, 16)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 300)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 98)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 32, 300)
Linear depth:1
input: (10, 16)
output: (10, 128)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64, 98)
output: (10, 64, 98)
relu depth:1
input: (10, 32, 300)
output: (10, 32, 300)
relu depth:1
input: (10, 128)
output: (10, 128)
Conv1d depth:1
input: (10, 64, 300)
output: (10, 96, 300)
Conv1d depth:1
input: (10, 64, 98)
output: (10, 96, 15)
Conv1d depth:1
input: (10, 32, 300)
output: (10, 64, 300)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 96, 300)
output: (10, 96, 300)
relu depth:1
input: (10, 96, 15)
output: (10, 96, 15)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64)
output: (10, 64)
Conv1d depth:1
input: (10, 96, 300)
output: (10, 128, 300)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 128, 300)
output: (10, 128, 300)
AdaptiveAvgPool1d depth:1
input: (10, 96, 15)
output: (10, 96, 1)
relu depth:1
input: (10, 32)
output: (10, 32)
AdaptiveAvgPool1d depth:1
input: (10, 128, 300)
output: (10, 128, 1)
AdaptiveAvgPool1d depth:1
input: (10, 64, 300)
output: (10, 64, 1)
unsqueeze depth:1
input: (10, 32)
output: (10, 32, 1)
cat depth:1
input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1)
output: (10, 320, 1)
squeeze depth:1
input: (10, 320, 1)
output: (10, 320)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
Linear depth:1
input: (10, 32)
output: (10, 5)
Linear depth:1
input: (10, 32)
output: (10, 5)
Linear depth:1
input: (10, 32)
output: (10, 5)
output-tensor depth:0
(10, 5)
output-tensor depth:0
(10, 5)
output-tensor depth:0
(10, 5)